A Component-Level Evaluation of Web Interface Vulnerability Under Simulated Colour Vision Deficiencies

Luke Vassallo
Institute of Information & Communication Technology
Malta College or Arts, Science & Technology
Corradino Hill
Paola PLA 9032
luke.vassallo.g52924@mcast.edu.mt

***Abstract*—Abstract goes here**
***Index Terms*—Keywords go Here**

I. Introduction

A. Description of Theme and Topic Rationale

Web accessibility is a fundamental pillar of inclusive digital design, ensuring that websites, applications, and online systems align with the POUR principles: Perceivable, Operable, Understandable, and Robust. These principles guide developers in creating interfaces that support users with visual, auditory, motor, cognitive, and perceptual differences [1]. As the web increasingly functions as a gateway to education, employment, communication, and services, inaccessible design risks limiting participation for a significant portion of the population [1], [2].

Accessibility evaluation extends beyond general compliance checking. Current frameworks emphasise the need for structured and measurable criteria that allow objective comparison across platforms. Petrie and Bevan highlight the importance of distinguishing accessibility from broader usability and user experience concerns, noting that accessibility should be evaluated through systematic and transparent methods [2]. This highlights the need for quantitative approaches capable of identifying measurable accessibility weaknesses within user interfaces.

An estimated 300 million individuals worldwide experience some form of Colour Vision Deficiency (CVD), including around 1 in 12 men and 1 in 250 women [3]. Because colour is widely used to communicate states such as alerts, success messages, navigation cues, and data visualisations, users with CVD may experience reduced clarity when interacting with colour-dependent interfaces.

Research has shown that simulated CVD conditions can influence how clarity and usability are perceived in web interfaces [3]. While prior studies provide valuable insight into colour perception and interface usability, many evaluate interfaces as a whole or focus on predefined colour schemes. Limited research has examined how individual UI component categories behave under simulated CVD conditions across multiple websites. This reveals a gap in structured component-level analysis capable of identifying patterns of contrast vulnerability.

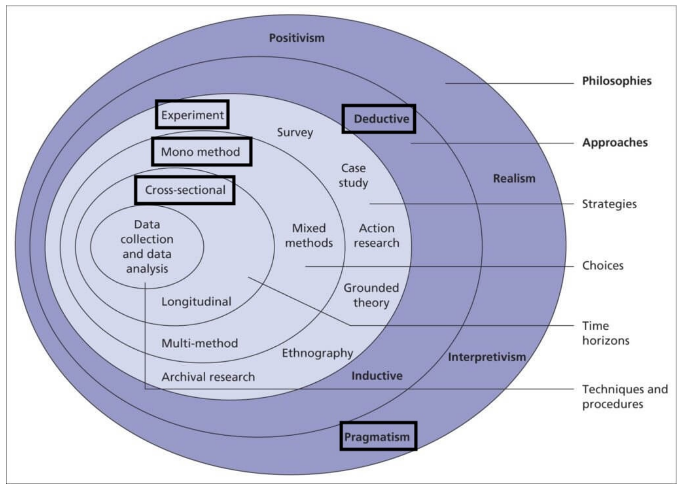

Fig. 1: Research onion illustrating philosophical and methodological positioning

B. Positioning and Research Onion

As shown in Figure 1, the Research Onion represents the layers of methodological decisions that guide the structure of this study.

This research adopts a philosophy of realism, viewing accessibility issues such as foreground–background contrast ratios as measurable phenomena that can be evaluated against WCAG thresholds. A deductive approach is used to test a predefined hypothesis regarding how UI components perform under simulated CVD conditions.

The methodological choice is mono-method quantitative, relying on measurable contrast ratios and pass–fail outcomes across predefined component categories. The research strategy follows an experimental design in which selected websites are analysed under normal vision and simulated CVD conditions using a consistent automated evaluation program. The independent variable is the simulated vision condition, while the dependent variable is the contrast performance of UI components. The time horizon is cross-sectional, as all websites are analysed at a single point in time.

C. Background to this Research Theme

CVD affects the ability to distinguish between certain colour ranges, most commonly within the red–green spectrum [4]. Because colour plays a central role in modern user interfaces, reduced colour differentiation can affect recognition of navigation elements, buttons, form indicators, alerts, and graphical components [3], [5].

The Web Content Accessibility Guidelines (WCAG) provide established contrast thresholds intended to improve accessibility. However, these measurements primarily evaluate static foreground–background contrast and may not fully capture perceptual differences experienced under simulated CVD conditions [3], [6]. This creates the need for structured evaluation methods capable of analysing contrast performance under different visual conditions.

D. Hypothesis

Simulated CVD conditions will reveal measurable differences in foreground–background contrast performance across user interface component categories, indicating patterns of accessibility vulnerability.

E. Research Questions

This study is guided by the following research questions:

- To what extent do user interface components across selected websites maintain acceptable foreground–background contrast thresholds under simulated CVD conditions?
- How does contrast performance vary across different user interface component categories when evaluated under simulated CVD conditions?
- Which UI components demonstrate the highest levels of contrast vulnerability when analysed across multiple websites?

F. Research Aim and Purpose Statement

The aim of this research is to evaluate how different user interface component categories perform in terms of foreground–background contrast when analysed under simulated CVD conditions.

The purpose of this experimental study is to investigate how simulated CVD conditions influence the contrast performance of UI components across selected websites in order to identify patterns of accessibility vulnerability.

The independent variable is the simulated vision condition (normal vision, protanopia, deuteranopia, and tritanopia), while the dependent variable is the measured foreground–background contrast performance of UI components evaluated using WCAG contrast thresholds. Control variables include the use of a consistent automated scanning script, identical contrast evaluation criteria, and the same set of analysed websites.

II. Literature Review

This chapter reviews research on web accessibility evaluation, focusing on methodologies used to identify accessibility issues and usability examination. This review focuses on automated evaluation, user testing, and hybrid approaches, critically comparing their strengths and limitations in UI accessibility testing. It also explores research on CVD, including simulation-based evaluation and colour design strategies. The chapter concludes by identifying methodological gaps that inform the proposed research.

A. Accessibility Evaluation in Web Interfaces

Web accessibility is essential as digital services expand across public, educational, and commercial sectors. The WCAG, developed by the World Wide Web Consortium (W3C), provide structured recommendations to ensure content is perceivable, operable, understandable, and robust [1].

WCAG defines accessibility through testable criteria that guide the design of interface elements such as navigation, forms, and interactive components. However, while WCAG provides a comprehensive framework, it does not mention neither specify how accessibility should be evaluated in practice. This has led to the development of multiple evaluation methodologies/tools, each addressing different aspects of accessibility assessment.

B. Approaches to Web Accessibility Evaluation

Accessibility evaluation methods are commonly classified into automated evaluation, expert inspection and specific audience user testing [7].

Automated tools analyse webpages against accessibility guidelines and efficiently detect structural issues such as missing alternative text or insufficient contrast. However, their rule-based nature limits their ability to interpret context or user intent, making them less effective for identifying usability-related issues.

In contrast, manual expert evaluation provides deeper analysis by allowing specialists to asses context and interaction design. While this approach identifies issues that automated tools miss, it lacks scalability and introduces a subjective approach to testing [7].

User testing offers the most direct insight into accessibility by observing real users interacting with interfaces [7]. Compared to automated and expert methods, it captures practical usability barriers more effectively. However, it is resource-intensive and difficult to generalise across different user groups.

Hybrid approaches attempt to combine these strengths by integrating multiple techniques. While more comprehensive, they also inherit the complexity and cost of each method, limiting their practicality for large-scale evaluation [7].

C. Evaluation Tools and Accessibility Metrics

Automated accessibility tools are widely used due to their efficiency, but their effectiveness varies significantly. Alsaeedi proposes evaluation frameworks using quantitative metrics to compare tools and webpages [7].

The Coverage Error Ratio (CER) measures how many accessibility violations a tool detects relative to others, while Web Accessibility Accuracy (WAA) evaluates overall accessibility based on detected errors. These metrics provide a structured basis for comparison.

However, despite offering quantitative insight, these metrics rely on the outputs of automated tools, which are inherently limited. Inconsistent error reporting, false positives, and undetected violations reduce reliability. This suggests that metric-based evaluation alone cannot fully represent accessibility quality.

D. Simulation-Based Accessibility Evaluation for CVD

Research has increasingly focused on accessibility for users with CVD, which affects the perception of colour differences in interfaces.

Simulation-based methods apply computational models to replicate different types of colour vision impairment. Jamil and Denes evaluate accessibility by applying CVD simulations to interface designs and analysing user responses [3]. Compared to traditional user testing, these methods enable scalable and controlled evaluation.

However, simulation-based approaches introduce limitations. They are often conducted using static screenshots and participants without CVD, which reduces ecological validity. As a result, simulated findings may not fully reflect real user experiences.

E. Colour Contrast and Accessible Colour Design

Foreground and background contrast is a critical factor in accessibility, as insufficient contrast proves to decrease usability and distinguishability for most users [1].

Sajek et al. analyse colour schemes using simulation and perceptual models such as CIELAB to quantify colour differences under CVD conditions [5]. In contrast, Troiano et al. focus on generating accessible colour palettes using optimisation techniques [8].

Whilst these two studies use similar approaches in terms of contributions for improved colour accessibility, both primarily operate at the level of global colour schemes. Consequently, they do not fully address how accessibility issues emerge within specific interface components or interaction states.

F. Critical Analysis and Research Gaps

The literature demonstrates that accessibility evaluation methods involve clear trade-offs. Automated tools provide efficiency but lack contextual understanding, whereas expert evaluation and user testing offer deeper insights but are difficult to scale. Hybrid approaches improve coverage but remain resource-intensive.

Similarly, research on CVD predominantly focuses on colour scheme evaluation and simulation-based analysis, with limited attention given to component-level accessibility. This creates a gap between theoretical colour optimisation and practical interface usability.

Furthermore, many studies assess accessibility at the page level rather than analysing individual UI elements(component-level). This limits the ability to identify recurring patterns of failure within components such as buttons, forms, or navigation systems.

These limitations indicate the need for methodologies that operate at a much more detailed manner . By combining colour vision simulation with structured evaluation frameworks, it becomes possible to analyse accessibility at the component level and identify specific vulnerabilities under CVD conditions.

G. Literature Map

Figure 2 presents a structured overview of the key concepts, methodologies, and relationships identified within the literature. The map illustrates how WCAG provides the foundational framework for accessibility, while various evaluation methods such as automated tools, expert inspection, and user testing are used to assess compliance.

It also highlights how colour vision deficiency (CVD) research extends accessibility evaluation through simulation-based approaches and colour design techniques. Furthermore, the map emphasises the limitations of existing approaches, including the lack of contextual understanding in automated tools and the limited focus on component-level analysis in both accessibility evaluation and colour research.

These relationships collectively reveal a clear research gap, which motivates the need for a more granular, component-level accessibility evaluation approach under simulated CVD conditions.

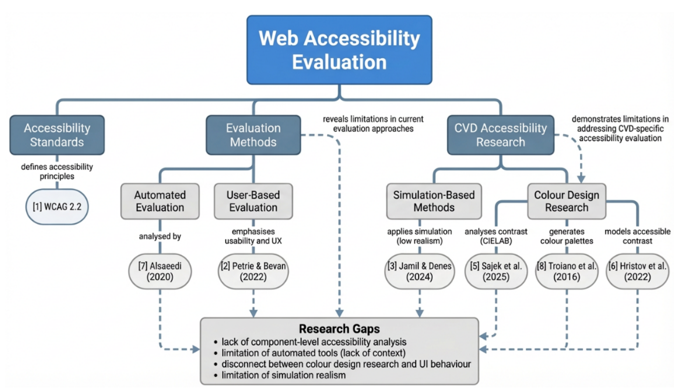

Fig. 2: Literature Map and Research Gaps

III. Research Methodology

Methodology goes here

IV. Findings and Results

Findings go here

V. Conclusion

Conclusion goes here

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